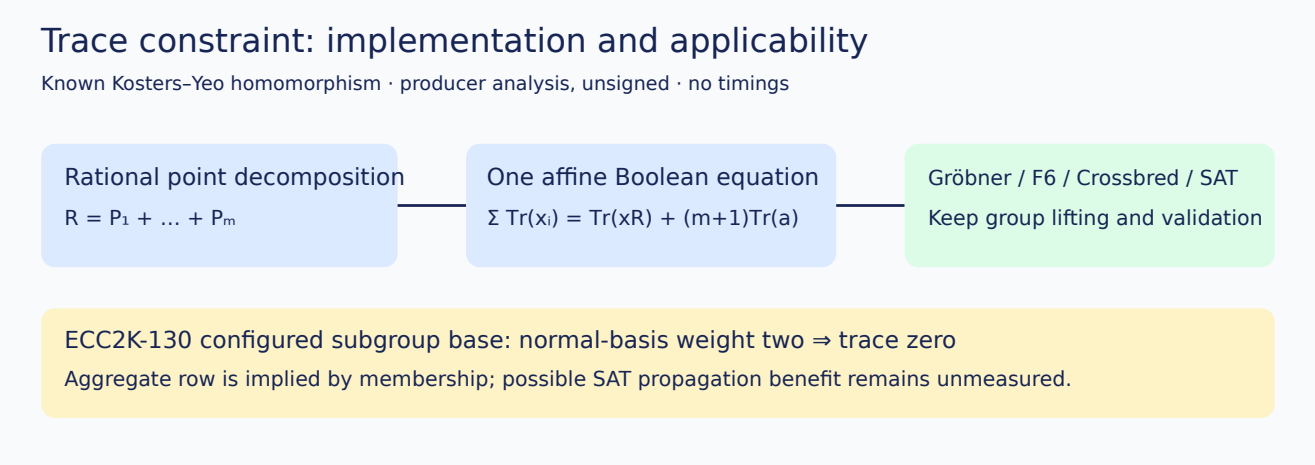
1. Binary trace constraints in point-decomposition solvers

2026-11-09 · Isogeny-labs/crypto · producer report, unsigned PCC standing.

The supplied screenshot is from Kosters and Yeo, *Notes on summation polynomials*, arXiv:1503.08001v3. The native SAT path already supplied its aggregate trace equation. This patch shares that construction with the Gröbner, F6 and Crossbred group-decomposition frontends. Explicit Gröbner and equation-export on/off controls retain a matched reference.



1.1. Mathematical contract

For the binary model $y^2 + xy = x^3 + a\,x^2 + b$, define delta(O) = 0 and delta(P) = Tr(x(P) + a). Proposition 4.2 of the paper gives a group homomorphism with kernel 2E. Thus $R = P_1 + \ldots + P_m$ implies $\Sigma\, Tr(x_i) = Tr(xR) + (m+1)Tr(a)$.

The equation is inserted before variable permutation and involves only summand coordinates. Zero equations are skipped; contradictions are kept; duplicate equations are not added. Full rational-point lifting remains mandatory. Raw summation-polynomial builders keep their original contract: Remark 4.8 warns that higher-polynomial algebraic roots need not satisfy the rational-point relation. These are implementation assertions awaiting independent signed verification.

1.2. ECC2K-130 applicability

The exact challenge is E_0: $y^2 + xy = x^3 + 1$ over $GF(2^{131})$, represented by $z^{131} + z^{13} + z^2 + z + 1$. It is different from the random ECC2-131 challenge. This study uses the published generator and target through BinaryCurve.certicom_challenge and the existing degree-83 E_0 fixture.

The fixed campaign configuration selects a subgroup normal-basis base of weight at most two. Trace is parity in every normal basis; subgroup points have trace zero on E_0. Hence its admitted nonzero coordinates have weight two, and the aggregate trace row is 0 = 0 on that domain. This is an applicability analysis, not a campaign runtime measurement.

With ambient SAT coordinates the row is implied by membership, rather than a literal zero polynomial. An explicit redundant XOR could still help propagation; that effect is unmeasured. Literal zero-row omission applies to a trace-zero linear coordinate basis.

The generic polynomial FieldStructure is exact only through field degree 64. That guard is preserved. Arithmetic checks at degree 83 and 131 do not establish a wide-field Gröbner solver or a full-DLP comparison.

1.3. Validation and remaining measurements

Native tests cover parity signs for a = 0/1 and m = 2/3/4, intermediate variable independence, zero and contradictory rows, duplicate suppression, trace-on/off agreement with exhaustive search, group lifting and exported variable order. Further checks use the published challenge

points and rank-check a normal basis in the actual challenge field, checking trace parity. Consult VALIDATION.md for the actual command outcomes.

No timing or speedup is reported. The required S3 experiment checkpoint cannot be pushed with the current session access; the matched EKS study, external solver comparisons, degree-83 full-DLP comparison and challenge campaign measurement remain pending. The planned controls and admission requirements are retained in PROTOCOL.md. No measurement changes the leaderboards or performance graphs.

1.4. Sources and reproducibility

Baseline source: e472218c8b14a770c38a1198bc2073edbc03d480.

Primary source: Kosters–Yeo, Proposition 4.2, Proposition 4.7 and Remark 4.8.

Repository sources: koblitz_index_calculus.rs (SAT and group-decomposition frontends); koblitz_groebner.rs (exactness guard); binary_ecc/curve.rs (challenge); docs/ic/params/ecc2k130-fixed.json and codegen/indexcalc_fixed.py (configured subgroup normal-basis recipe). README.md records precise paths, requirements and all pending deliverables.

Knowledge-base fallback: no MCP tools loaded; scripts/research-kb status exited 127 because no executable was installed. No novelty claim is made.

1.5. Final local checks

Five focused trace tests and six existing decomposition/status regressions pass in the corrected diagnostic build: release profile, package crypto opt-level=0. These are unsigned producer checks, not admitted timing or independent verification. The original optimized run exposed the retained two-torsion case: three copies of (0,1) sum to (0,1), but an intermediate sum is infinity. Chained frontends now report such targets unsupported rather than refuted. No missing infinity branch is claimed implemented.

The earlier whole-library run was interrupted after about 13 minutes and was already red; isolated failures identify missing GPU source and a historical session record. Other failures are unclassified. Ordinary optimization rerun of the final source, full-suite validation, signed verification and performance measurements remain pending. Exact command outputs are retained in checks/ and VALIDATION.md.